

PUGAZHENDHI S

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PROFILE

Software Development Engineer with strong expertise in Java, Python, Data Structures and Algorithms, and backend API development using FastAPI. Experienced in building scalable RESTful services, implementing JWT-based authentication, optimizing database queries (MongoDB, PostgreSQL), and deploying applications using Docker and AWS. Passionate about writing clean, efficient, and production-ready code with strong problem-solving skills.

EDUCATION

B.Tech in AI and Data Science, St. Joseph's College of Engineering, Chennai

CGPA: 8.80 (current)

TECHNICAL SKILLS

Programming Languages: Java, Python

Frontend Technologies: HTML5, CSS3, JavaScript, TypeScript, React, Angular

Backend Technologies: FastAPI, RESTful APIs, JWT, OAuth2

Databases: MySQL, PostgreSQL, MongoDB, Oracle SQL

Cloud & DevOps Tools: Docker, Kubernetes, Google Cloud Platform, Amazon Web Services, Git, GitHub, GitHub Actions

Generative AI & AI Frameworks: LangChain, LangGraph, CrewAI

Testing Frameworks: JUnit 5

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, Full Stack Development

EXPERIENCE

Machine Learning Intern, Bigbucks Innovation Limited

- Built a CNN model for digit classification using the MNIST dataset.
- Developed a breast cancer prediction system using SVM and Logistic Regression.

Machine Learning Intern, National Small Industries Corporation Limited

- Developed ML models for data analysis and industrial predictions.
- Assisted in integrating ML techniques into business processes.

PROJECTS

AI-based Virtual Mock Interviewer – Gemini API, VAPI, Firebase

- Developed an AI-powered mock interview application simulating real-time voice-based interviews.
- Integrated Gemini API for natural language understanding and VAPI for voice interaction.
- Used Firebase for real-time user data management and authentication.

Task Management System – FastAPI, Python, OAuth2, JWT, MongoDB, PostgreSQL

- Designed and deployed scalable RESTful APIs handling CRUD operations for 500+ simulated users.
- Implemented secure OAuth2 authentication with JWT-based role authorization.
- Optimized MongoDB indexing, reducing API response time by 35%.
- Designed hybrid database architecture (MongoDB 80%, PostgreSQL 20%) for performance & consistency.
- Containerized application using Docker for deployment readiness.

Customer Support Email Agent – LangGraph, FastAPI, PostgreSQL, RAG

- Developed an AI-powered email automation system that fetches incoming emails, classifies intent, and processes customer queries intelligently.
- Implemented Retrieval-Augmented Generation (RAG) to generate context-aware response drafts using customer support documentation.
- Automated bug ticket creation in PostgreSQL for issue-related emails with database checkpointing support.
- Designed a human-in-the-loop workflow for handling high-complexity or critical intent emails.
- Built modular backend services using FastAPI and orchestrated agent workflows using LangGraph.

AI-Powered PPT Evaluator | Python, Streamlit, LangChain, Gemma 3

- Developed a Streamlit-based evaluation tool powered by the Gemma 3 language model to assess and rank PowerPoint presentations on 7 key criteria with real-time scoring.
- Integrated parallel processing for batch evaluation and implemented dynamic visualizations to display weighted results for better insights.

Library Management System – Java, JDBC, MySQL, OOPs

- Developed a Library Management System using Java with Object-Oriented Programming principles.
- Integrated MySQL database using JDBC for managing books, members, and transactions.
- Implemented features such as book issue/return, search functionality, and overdue fine calculation.

CERTIFICATIONS

Certified System Administrator – ServiceNow

Database Management System – NPTEL

Introduction to Machine Learning – NPTEL

Python for Data Science – NPTEL

Technology & Transformation Launchpad Program – PwC

ACHIEVEMENTS

SQL Gold Badge – HackerRank

Problem Solving Silver Badge – HackerRank

Python Silver Badge – HackerRank

100 Days Badge - SkillRack

100 Days Badge - LeetCode